Drug Review Classification with Custom LSTM Cells

Universite Internationale de Rabat - Advanced NLP - 2026/2027
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1. Objective and principal findings

We compare Vanilla LSTM, Coupled Input-Forget Gate LSTM (CIFG), and Working Memory Connections (WMC) on rating-derived drug review classification. All cells follow the course appendix and use manual PyTorch nn.Linear transformations. The validation-selected Vanilla configuration achieves 69.40% accuracy and 0.5583 Macro-F1 on 48,257 cleaned test reviews. This is one reduced-data, single-seed experiment, rather than a general ranking of LSTM designs.

2. Dataset and target definition

The supplied Drugs.com CSVs contain 161,297 training and 53,766 test rows. Only review text enters the classifier. The rating constructs the label and never enters the input. Drug name, condition, date, usefulCount, and uniqueID are excluded from predictive features. The three-class mapping follows the supplied draft report and is a project choice, not a threshold prescribed by the guidelines.

Rating	Class	Meaning
1-4	Negative (0)	Lower reported satisfaction
5-6	Neutral (1)	Intermediate reported satisfaction
7-10	Positive (2)	Higher reported satisfaction

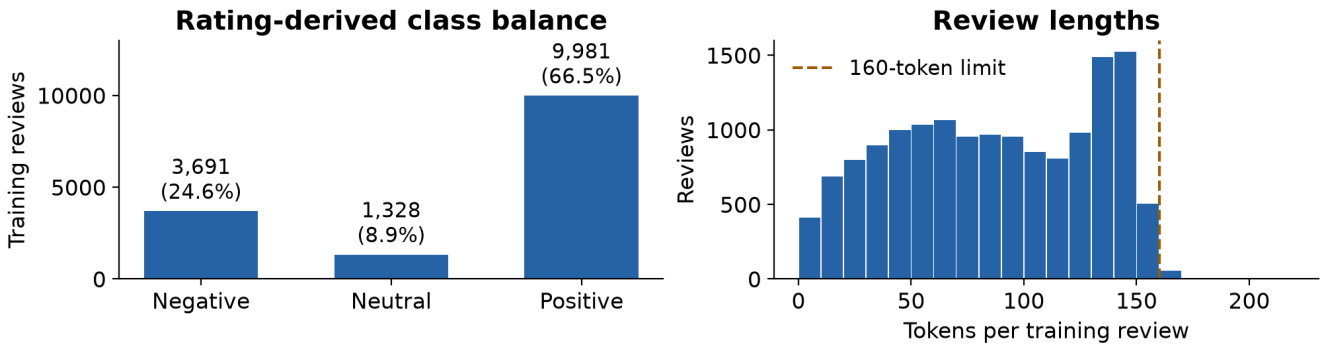


Figure 1. The 15,000-review training subset preserves the source class proportions. Positive reviews dominate (66.54%), while Neutral accounts for only 8.85%. Median length is 86 tokens and the 95th percentile is 148 tokens.

3. Quality checks and duplicate leakage

Review IDs are unique in both source files. Condition is missing in 899 training and 295 test rows, which does not prevent a text-only task. After HTML decoding, case normalization, and whitespace cleanup, 49,012 training rows repeat an earlier review text. The test file also contains 5,504 repeated text rows. Identical texts crossing partitions could reward memorization, so we remove them before modeling.

4. Preprocessing and shared data protocol

Each source file is cleaned independently. We discard invalid or token-empty reviews, remove all texts with conflicting three-class labels within that file, and keep one copy of each remaining text. We then exclude every cleaned training text that appears anywhere in the raw test file, regardless of its rating. This removes 32,111 candidate training rows. Test labels do not enter vocabulary construction, weighting, optimization, or model selection.

Stage / split	Rows	Selection rule
Eligible source training	80,159	After cleaning and excluding test texts
Training pool / validation pool	64,127 / 16,032	Stratified 80/20 split, seed 42
Executed training / validation	15,000 / 3,000	Fixed stratified subsets, seeds 42 / 43
Evaluation test set	48,257	All unique valid, unambiguous test texts

The guidelines permit reduced training data when compute is limited. All 12 runs use these exact same partitions, saved by ID and text hash. The 12,000-entry vocabulary, including PAD and UNK, is fitted only to the executed training subset. Tokenization retains negation and contractions. Each review contributes its first 160 tokens, with dynamic batch padding and true-length state masking. Only 0.46% of training and 0.52% of test reviews are truncated. Test unknown-token rate is 1.05%.

5. Recurrent architectures and appendix equations

For each gate q, define z_q = W_xq x_t + W_hq h_(t-1) + b_q. Vanilla uses i = sigmoid(z_i), f = sigmoid(z_f), o = sigmoid(z_o), g = tanh(z_g), c_t = f*c_(t-1) + i*g, and h_t = o*tanh(c_t). Multiplication is elementwise. We concatenate gate matrices for efficiency while preserving the same nn.Linear operations and equations.

CIFG removes the independent input gate and sets i = 1 - f. WMC adds tanh(W_ci c_(t-1)) to z_i, tanh(W_cf c_(t-1)) to z_f, and tanh(W_co c_t) to z_o. The output gate therefore reads the updated memory c_t. WMC uses full learned memory projection matrices, with no extra memory bias. All models share an embedding, the final valid hidden state, dropout, and a three-logit linear output.

Cell	Recurrent parameters	Total parameters
Vanilla	12,416	780,515
CIFG	9,312	777,411
WMC	15,488	783,587

For embedding dimension E=64 and hidden size H=32, Vanilla has 4H(E+H+1) recurrent parameters, CIFG has 3H(E+H+1), and WMC adds 3*H^2 to Vanilla. CIFG reduces the recurrent cell by 25%. The shared 768,000-parameter embedding dominates total size, so the reduction in the complete classifier is much smaller.

6. Training controls and hyperparameter study

Each run starts from seed 42 with 64-dimensional embeddings and a 32-dimensional hidden state. Embeddings start from a zero-mean normal distribution (standard deviation 0.1), with a fixed zero padding row. Input gate blocks use Xavier initialization, recurrent gate blocks use orthogonal initialization, and the forget-gate bias starts at 1. Dropout is 0.2 on embeddings and the final representation. The loss is class-weighted cross-entropy, with class weight N/(3*n_class) computed from training labels only. All runs clip the global gradient norm at 1.

The fixed configuration uses AdamW, learning rate 0.003, batch size 256, and weight decay 0.01. Each of the other settings changes one factor: learning rate 0.001, batch size 128, or Adam instead of AdamW. Weight decay remains 0.01, so the optimizer comparison includes their different regularization semantics. Each architecture receives the same four-setting search budget, up to six epochs, with early stopping after three epochs without validation Macro-F1 improvement. Checkpoints maximize validation Macro-F1.

Setting	Vanilla val F1	CIFG val F1	WMC val F1
fixed	0.5425	0.5558	0.5571
lr_0.001	0.5576	0.5652	0.5447
batch_128	0.5657	0.5547	0.5556
adam	0.3389	0.2663	0.2856

Table 2. Best validation Macro-F1 for each run, from actual epoch histories. The checkpoint epoch and elapsed training time are saved with each experiment. The same maximum epoch budget does not imply identical gradient update counts when batch sizes differ.

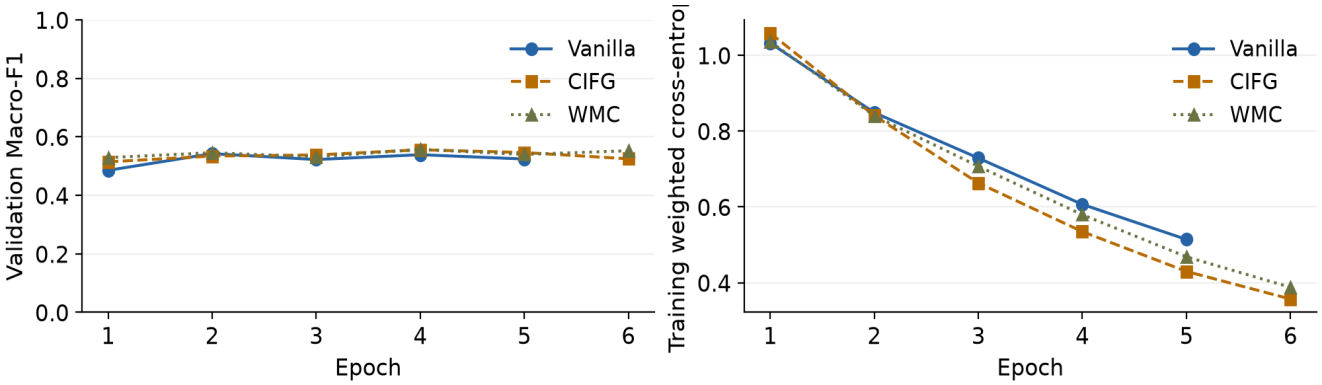


Figure 2. Fixed-setting learning curves. Left: validation Macro-F1. Right: weighted training loss. Training loss and unweighted validation loss use different reductions and should not be compared as equivalent quantities.

Runs execute on CPU with two PyTorch threads (PyTorch 2.15.0+cpu, Python 3.11.15). Timing includes training and per-epoch validation, but excludes preprocessing and final test inference. Exact package versions, source hashes, split manifests, configurations, checkpoints, and raw prediction files support reproduction.

7. Held-out evaluation and model comparison

Macro-F1 is the primary measure because it weights each class equally despite imbalance. Accuracy, per-class precision/recall/F1, confusion matrices, and computational cost provide complementary evidence. We freeze each architecture's validation winner before test scoring. Fixed-setting results isolate the cell change; selected-setting results compare equal-budget validation searches. No model or checkpoint is chosen by test performance.

Model	Fixed acc.	Fixed F1	Selected setting	Selected acc.	Selected F1
Vanilla	65.80%	0.5501	batch_128	69.40%	0.5583
CIFG	67.78%	0.5484	lr_0.001	70.50%	0.5650
WMC	70.41%	0.5583	fixed	70.41%	0.5583

A training-majority Positive classifier scores 65.96% accuracy and 0.2650 Macro-F1. Neural results should be judged against both, since high accuracy alone can hide minority-class failure.

Selected model	Negative P / R / F1	Neutral P / R / F1	Positive P / R / F1
Vanilla	0.713 / 0.542 / 0.616	0.171 / 0.374 / 0.235	0.855 / 0.796 / 0.824
CIFG	0.625 / 0.675 / 0.649	0.184 / 0.297 / 0.227	0.872 / 0.772 / 0.819
WMC	0.652 / 0.619 / 0.635	0.172 / 0.291 / 0.216	0.858 / 0.793 / 0.824

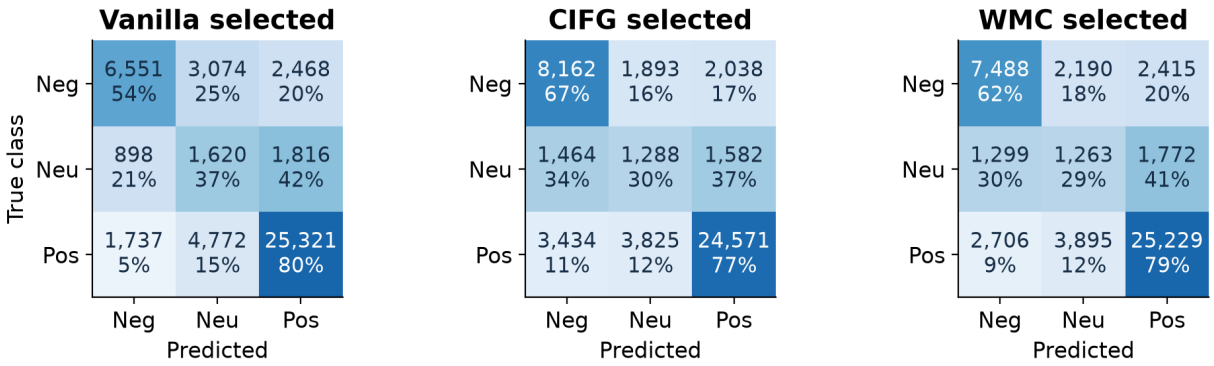


Figure 3. Selected models on the same test set. Counts and row percentages show which true classes each model confuses. Support: 12,093 Negative, 4,334 Neutral, and 31,830 Positive reviews.

Architecture	Fixed min / epochs	Search total (min)	Selected epoch
Vanilla	1.11 / 5	4.61	2
CIFG	1.21 / 6	5.01	5
WMC	1.68 / 6	6.62	4

8. Interpretation, errors, and limitations

At fixed settings, CIFG changes test Macro-F1 by -0.17 percentage points relative to Vanilla, while WMC changes it by +0.82 points. CIFG saves 3,104 recurrent parameters, corresponding to only 0.40% of the full Vanilla classifier. These results measure one initialization and one training subset. Small performance differences cannot establish architectural superiority without repeated seeds and a comparison uncertainty analysis.

The optimizer study yields validation Macro-F1 ranges of 0.266-0.339 for Adam and 0.543-0.557 for fixed AdamW. Weight decay 0.01 has different semantics in these optimizers. The weak Adam runs therefore describe this regularization setting, rather than proving that Adam is generally unsuitable. A follow-up should tune optimizer-specific regularization on validation data.

CIFG trades independent write control for fewer recurrent parameters. WMC gives gates direct access to cell memory at added cost. Their observed predictive results must be considered alongside these parameter counts and measured timing, rather than assuming that the more elaborate mechanism will perform better. The selected overall candidate follows validation Macro-F1, even if test rankings differ.

For the overall validation-selected candidate, Neutral is the weakest class (F1 0.235). This class-level result explains why overall accuracy must be read alongside Macro-F1. The confusion matrices identify the relevant destination classes without attributing the mistakes to an unverified linguistic cause.

Illustrative Vanilla error (review ID 215892): true Neutral, predicted Negative. Excerpt: "I've had the copper coil for about 3 months now. I was really excited at the thought of not taking hormones. I'm good with pain however I nearly fainted with insertion, couldn't belive how painful it was; the doctor did ..."

Illustrative Vanilla error (review ID 208087): true Negative, predicted Positive. Excerpt: "4 days in on first 2 weeks. Using on arms and face. Put vaseline on lips, under eyes and in nostrils to protect from cream. So far no reaction at all. I know I have many pre cancer and thought I would light up like a Chr..."

These deterministic examples illustrate individual mistakes and do not prove a systematic cause. Saved error-slice tables compare all reviews, a heuristic negation subset, reviews longer than 160 tokens, and Neutral reviews. Subset Macro-F1 can include absent classes, so the Neutral-only slice is interpreted through its accuracy (Neutral recall), not its three-class Macro-F1.

The principal limitations are the small training subset, a compact hidden state, a limited search, and one seed. Ratings are a noisy proxy for textual sentiment. Our strict text-deduplicated population differs from the raw official split, so its scores are not directly comparable with published results on all raw rows. Exact text deduplication does not detect paraphrases, shared templates, patient identity, or all potential group leakage. Random source splits do not test temporal or unseen-drug transfer.

9. Improvements and conclusion

Next experiments should retain the frozen test protocol while using the full eligible training pool, multiple seeds, and a broader validation search. Pretrained embeddings or attention may improve representations, while additional class-balance strategies and error annotation may clarify Neutral performance. Longer context should be tested only where the small truncated subset warrants its extra cost. The delivered project establishes a controlled comparison of the three required custom cells with measured predictive and computational results.

References and reproducibility

[1] Y. Moukafih. Lab Project on Recurrent Neural Networks, Advanced NLP, UIR, 2026/2027, pp. 1-2. Supplied guidelines and appendix.
[2] Supplied Drugs.com training and test CSVs. Course dataset reference: archive.ics.uci.edu/dataset/462.
[3] K. Greff et al. LSTM: A Search Space Odyssey. arxiv.org/abs/1503.04069.
[4] F. Landi et al. Working Memory Connections for LSTM. arxiv.org/abs/2109.00020.
[5] PyTorch 2.15 Adam and AdamW API documentation: docs.pytorch.org/docs/2.15/generated/torch.optim.Adam.html and torch.optim.AdamW.html.

PyTorch repository: https://github.com/isma-ii1/drug-review-custom-lstm
Results: artifacts/validation_results.csv, test_results.csv, evaluation/, error_slices.csv. Split and preprocessing audit: data/processed/data_audit.json and split_manifest.csv.